

1. Data Processing

Image Preprocessing:

All images were resized to 96x96 pixels and normalized to the range [0, 1].

Data Augmentation:

Applied **random horizontal flipping** (`tf.image.random_flip_left_right`) to the training set to increase data diversity and model robustness.

Train/Validation Split:

Used an 85:15 split, ensuring a balanced class distribution via **stratified sampling** to guarantee the validation set accurately reflects the population.



2. Model Architecture and Modification

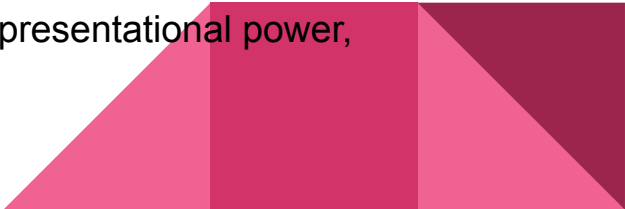
Base Architecture:

An **Enhanced Residual Convolutional Network (ResNet)** was used, employing **Residual Blocks** to ensure stable training in deeper layers.

Modifications for Performance:

- **Increased Depth:** Added **extra Residual Blocks** at the 64 and 128 channel stages (making it two blocks per stage).
- **Feature Enrichment:** Added a subsequent **256-channel ConvBlock** to enhance high-level feature extraction.
- **Dimension Reduction:** Added an **extra MaxPooling2D layer** and used **GlobalAveragePooling2D** to compress spatial information efficiently.

Performance Impact: These changes successfully boosted the model's representational power, contributing directly to the final 99.45% validation accuracy.



Model Structure

Stage	Output Shape	Operation Sequence	Notes
	(96, 96, 3)	Input Layer	
1	(48, 48, 32)	ConvBlock(32) → MaxPooling	Initial feature extraction.
2	(24, 24, 64)	ConvBlock(64) → ResBlock(64) ×2 → MaxPooling	Increased depth for 64 channels.
3	(12, 12, 128)	ConvBlock(128) → ResBlock(128) ×2 → MaxPooling	Further depth increase and dimension reduction.
4	(12, 12, 256)	ConvBlock(256)	Highest channel count before classification.
-2		GlobalAveragePooling → Dropout(0.5) → Dense(2, Softmax)	Parameter efficient head with strong regularization.



3. Model Training

1. Training Efficiency (Speed)

- **Mixed Precision:** Implemented the `mixed_float16` policy, enabling the GPU to use half-precision for most computations.
 - **Benefit:** **Significantly reduced single-epoch training time** (accelerating the overall process) while maintaining numerical stability.

2. Training Stability (Accuracy)

- **Smart Learning Rate Scheduling:** Used `ReduceLROnPlateau` to monitor `val_loss`. The learning rate was automatically reduced by 50% if validation loss plateaued for 3 epochs, aiding precise convergence.
- **Batch Size:** Adjusted `BATCH_SIZE` to **32** for potentially more robust gradient updates.

3. Regularization (Generalization)

- **High Dropout Rate:** Increased the **Dropout rate to 0.5** in the classification head.
 - **Benefit:** Provides strong regularization to effectively prevent the model from overfitting the large training dataset.
- **Early Stopping:** Used **EarlyStopping** (patience 5, monitored `val_loss`) to automatically terminate training when performance leveled off and restore the best weights, preventing overfitting.

Kaggle Snapshot

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